

Audubon Core Enhancement for 3D Imagery

Developed by

TDWG Audubon Core Task Group for 3D enhancement:

Doug Boyer – Duke Univ. (Director/Founder MorphoSource.org)

Jon Blundell – Smithsonian (3D Program Officer)

Gary Motz – Indiana Geological & Water Survey (Asst. Director for Information Services)

Adam Rountrey – Univ. of Michigan Mus. of Paleontology (Collection Manager/3D Specialist)

Rebecca Snyder – Smithsonian NMNH (Head, Informatics and Data Science Center)

Jocelyn Triplett - Duke Univ., MorphoSource.org

Kate Webbink – Field Museum (Information Systems Specialist)

Julie Winchester – Duke Univ. (Product Manager/Lead Dev. MorphoSource.org)

Link to folder containing this and other documents linked here:

<https://drive.google.com/drive/folders/10Z43lqpeKlKaeOuRyLtEJBzi1wmlSYXn?usp=sharing>

Contents

[Proposed Terms list](#)

[Use of terms: volumes, polygonal meshes, and point clouds](#)

[Use of terms: scale, dimensions, and resolution](#)

[I. Terms communicating file type and imaging modality](#)

[II. Terms communicating resolution via pixel/voxel \(raster\) or point cloud/mesh scale:](#)

[III. Terms communicating dimensions or pixel/data counts of image](#)

[IV. Terms communicating additional information about object](#)

Audubon Core Enhancement for 3D Imagery

Proposed Terms list

The full list of proposed terms is given [here](#). Please continue to the remainder of the document for discussion supporting these choices, as well as definitions and usage notes for each.

Use of terms: volumes, polygonal meshes, and point clouds

There exists a diversity of techniques (aka: [modalities](#)) that are commonly used to generate digital 3D data. These modalities result in data that tends to fall into one of two broad categories: data that convey information on volume (including internal structure) and data that convey information on the external surface. Volumes are typically encoded as raster data, while surfaces are encoded as either point clouds or polygonal meshes. Volume data, point clouds and polygonal meshes are the three basic classes of 3D data. While these different image classes encode different kinds of 3D information and sometimes encode the same information in technically different ways, we believe they share some of the same basic, categorical metadata requirements.

Some kinds of raw data necessary to produce volumes or 3D surface images cannot be visualized as 3D images without further processing. The raw data we have in mind here includes 1) CT projection series that must be “reconstructed” into an “image slice stack” in order to be visualized as a volume, and 2) photographic collections that are processed to make polygonal mesh photogrammetry models. We think these can be considered “3D resources” in the sense that they contain all information necessary for reconstructing a 3D dimensional image. Access to these raw data file collections is essential for verifying the quality and accuracy of the volumes and surface files they underlie. Because of this, we deemed it necessary to include these raw “3D resources” within the scope of this vocabulary enhancement.

The volume class of 3D images consist of raster data. Specifically, volume data are arrays of discrete voxels that can take on a range of values representing visual or physical qualities (gray scale or color). Standard 2D digital photographs are raster data arranged in two dimensions (X and Y). Raster 3D data is formed either by voxels arranged in three dimensions (X, Y, and Z) or by a collection of individual 2D images where the ordering of the image collection may or may not specify an additional Z dimension. Raster data produced by CT/MRI formats can exist as either of the above. “3D pixels” are commonly called “voxels”. Voxels may be cubic or anisotropic in their scale/resolution.

Polygonal meshes (or just “meshes” for short) are considered broadly to include any 3D data that is encoded by one or more polygons, typically triangles, represented in 3D space using a coordinate system with X, Y, and Z dimensions. In addition, when considering the traditional use

Audubon Core Enhancement for 3D Imagery

of the term “polygonal mesh”, mesh files also include information on polygonal connectivity of points, (i.e., which points share edges). When edges connecting points are visualized as lines on a 3D model, the model has the appearance of being composed of a “mesh-like” fabric, which is where the term “mesh” comes from. When multiple points share a set of edges and enclose a polygon, it also makes sense to call the points “vertices” of the polygon. Thus, points of a mesh file are often referred to this way.

Some scanning modalities are not able to collect data on connections between points and thus natively produce “point cloud” data, which is just a list of XYZ coordinates. Furthermore, information on edges can be removed from a given mesh file and the file can be saved without this information as a point cloud.

Points (i.e., vertices), edges, and/or polygonal faces may also have additional information associated with them as part of the image data, including tangential or normal vectors, UV coordinates, color information, or others. Mesh imagery can be produced by line-of-sight scanners or as derivatives from photogrammetry or CT/MRI image series.

The committee debated whether “texture maps” should be considered 3D resources. Unlike photograph collections for photogrammetry models and projection series mentioned above, texture maps do not contain information necessary to reconstruct a 3D scene and by this criteria we do not consider them a formal type of 3D resource. However, they provide essential information that is only meaningful in the context of an associated mesh model. In the sense that “texture maps” are virtually meaningless data without reference to an associated model and because many mesh models cannot be fully rendered as intended without reference to a texture map, we recommend that data producers package texture maps with the point clouds and polygonal meshes that reference them.

Use of terms: scale, dimensions, and resolution

Some of the terms proposed below convey information on the resolution of 3D data while others aim to report the dimensions of the imagery. Unfortunately, the term “resolution” is commonly misused to describe dimensions (as counts of pixels) comprising an image (e.g., your computer screen “resolution”) and not how clearly or finely *resolved* those pixels are relative to real space. **Dimensions** refers specifically to *counts* of pixels, points, or other discrete data packets that comprise a total image. For raster data, a pixel count is required for each axis. For 2D raster data, this means pixel counts along X and Y axes. For 3D raster data (the volume class of data), pixel counts are required along X, Y, and Z axes (again, 3D pixels are commonly called “voxels”). Dimensions can be conceived of as the “amount of information or data” in a file and usually correlates with file size when controlling for file format. Thus changing the dimensions of a file by averaging or splitting pixels will have the effect of decreasing or increasing the file size, respectively. When dimension changes are executed correctly, the resolution of the file should

Audubon Core Enhancement for 3D Imagery

also be changed, however this will not always happen automatically like changes in file size. Two images or meshes with the same dimensions and same file size may be of higher or lower resolution depending on the real world scale of pixels or the real world distances between points, respectively.

I. Terms communicating file type and imaging modality

Currently, ac:subtype has a controlled vocabulary specifying a variety of resources. We propose the addition of “Digital 3D Resource” to the list of terms. There are a variety of Digital 3d Resources created in a variety of ways as discussed above. We propose two terms with controlled vocabularies to specify this information. “3D Resource Type” and “Image Capture Modality”.

Specifically we are planning to handle information on “3D Resource Type” and 3D data “Image Capture Modality” with hierarchical SKOS concepts in RDF:

<https://www.w3.org/2004/02/skos/intro>. Proposed ontologies are [here](#).

URI, Title & Definitions:

ac:3DResourceType: The type of file or the way the 3D information is encoded. It is not the same as format. An example of a type would be “mesh” which could be saved in a variety of formats (.ply, .obj, .glb, etc.). Neither is “type” redundant with “modality”. Mesh files can be produced by a variety of modalities. Uses SKOS concept in RDF (see linked proposed ontology above).

ac:resourceCreationTechnique: This is an already existing term (<http://rs.tdwg.org/ac/terms/resourceCreationTechnique>) that can be used to specify imaging/sensor and technology used to generate a 3D resource. We have created an ontology using SKOS concept in RDF for 3D modalities that can be used in this field (see linked proposed ontology above).

II. Terms communicating resolution via pixel/voxel (raster) or point cloud/mesh scale:

This class of terms reflects the resolution of raster data and scale of point and mesh data. Because of lacking standards for how resolution and scale are encoded, it is by far the most important set of extrinsic metadata for curators of 3D imagery to consider. To put it a different way, whereas it is trivial to extract the intrinsic dimensions of a data file at any time, the resolution of the imagery may not be intrinsically encoded in an unambiguous way. Therefore, without access to the following ‘scale-related’ metadata, it is possible that the associated 3D data will be unusable or used improperly. These terms describe the real world distances between either pixels/voxels in raster data or points/vertices in point clouds and meshes. In volumetric raster data, when pixel scale is present along with overall pixel dimensions, it is possible to calculate the real world dimensions of an object or a feature captured in an image/slice.

1. *Raster data*

Audubon Core Enhancement for 3D Imagery

For traditional 2D photographic raster images, the standard approach to conveying resolution has been to use “pixel density” terms. Specifically, pixels-per-inch (PPI) or dots-per-inch (DPI) have been used. EXIF treats these terms interchangeably and records pixels-per-inch values in the XResolution and YResolution fields. DPI values more properly refer to the number of dots capable of being physically printed by an Inkjet or other document printer.

It is important to emphasize that pixel scale or density for 2D digital photographs of 3D objects serve primarily to convey the “real world” size and sampling density of the image itself (when used in combination with image dimensions) and says little-to-nothing about the size and sampling density of 3D objects captured in the photograph. If there are several objects at different distances from the camera in an image, each would technically have a different pixel scale/density and resolution in the same photograph. Only in special conditions in which there is a very uniform distance of all objects from the camera and in which a scale bar is included and used to calculate “real world” resolution would PPI of the photograph and the objects in it coincide.

Alternatively, the standard convention for CT/MRI raster data has been to express image resolution as the linear distance from the center of one pixel to another along the X, Y, or Z axes. These pixel spacing measurements are the inverted counterpart to a document scanners’ PPI or DPI, and we prefer them here for a number of reasons. Primarily, we prefer them because they are already the community standard in use by some 3D formats (e.g., DICOM). However, we also feel use of “pixel spacing” helps emphasize the fundamentally different way resolution information can be used in 2D versus 3D raster images. As noted above, in 2D photographs, pixel scale allows understanding of the “real world” size of the image itself (e.g., if it were printed out on paper) when it is combined with dimension information. However, because real photographs are never fully orthographic, pixel scale of the image should not be used to estimate the “real world” size of objects in the image. Of course, scale bars can be used to adjust the pixel scale in a local part of the image in order to estimate the size of an object next to the scale bar, but this pixel scale will usually be different from the pixel scale that describes the “real world” size of the image. It will also probably be inaccurate for objects farther away from the scale bar in the image. The only exceptions here are special conditions where the photograph is of a very flat object on a flat surface that fills the entire field of view of the image (like a herbarium sheet). These conditions will approximate an orthographic image. **In contrast to pixel scale of 2D photographs, pixel scale of raster images that are part of a volumetric data set (when combined with image dimensions) should always convey both the “real world” size of the image itself and the “real world” size and resolution of any and all objects in the image.** Finally, even for circumstances in which PPI can be considered a reliable pixel-to-real-unit density measure, PPI emphasizes the number of pixels that may be represented in an **inch**, and so at a glance it is valuable for printing use cases but less valuable for individuals interested in how finely distinguished small features of an object may be in a

Audubon Core Enhancement for 3D Imagery

3D scan. Because of all this, pixel spacing more appropriately communicates the most essential and important information for biodiversity 3D scanning use cases than existing pixel density terms.

To indicate pixel spacing along X and Y axes, the DICOM standard uses “pixelSpacing (0028,0030)¹”. However, the values of this term are in the form of “row spacing/column spacing”, which is overly compact for flexible machine and human readability standards. Therefore, we recommend two separate terms to specify pixel spacing along X and Y axes respectively.

For volumetric raster data (e.g., that produced by CT or MRI) spatial pixel scale must be specified along the Z axis as well. We recommend borrowing and modifying the DICOM standard term for Z-axis slice spacing: SpacingBetweenSlices (0018, 0088). DICOM images also include a similar term called SliceThickness (0018,0050), but we do not consider this term appropriate for our purposes here. This is because SliceThickness describes the real-world Z-axis length that is averaged in the pixel values of a single slice. It is possible for slice spacing to be greater, equal to or less than slice thickness. When slice spacing is less than slice thickness, the pixel values of adjacent slices will be influenced by overlapping regions of the object and non-independent. More importantly using slice thickness in place of slice spacing to visualize a CT scan would result in a distorted display. Fundamentally, all this means is that SliceThickness should not be used as an indicator of z axis pixel scale, that it is not as critical for correctly interpreting the 3D form of a scanned object, and that it is instead more relevant for describing how slices were created or what slices represent. SpacingBetweenSlices provides more information on scaling and is the better term for conveying resolution. Yet per the DICOM standard, SpacingBetweenSlices must be encoded using millimeters as the associated unit. **We recommend a custom term, borrowed but distinct from SpacingBetweenSlices in order to have greater flexibility with regard to unit, and to in turn better support more varied 3D raster imaging modalities.**

URI, Title & Definitions:

ac:pixelXSpacing: The linear distance between each pixel in the X direction with units specified by associated Library of Congress bibframe term, Unit.

ac:pixelYSpacing: The linear distance between each pixel in the Y direction with units specified by associated Library of Congress bibframe term, Unit.

ac:pixelZSpacing: The linear distance between each pixel in the Z direction with units specified by associated Library of Congress term bibframe term, Unit. Usage note: For series of dicom images representing a volume, it is possible to have variable spacing between slices. In such cases pixelZSpacing should be left empty.

¹ The numbers following dicom terms are the codes for these terms in the dicom metadata standard.

Audubon Core Enhancement for 3D Imagery

2. *Point clouds and Mesh data*

Point cloud and Mesh resolution - or how finely spatially resolved the data is - can be described by specifying the units of the 3D coordinate system that is used to encode vertex/point position. This is equivalent to the pixel spacing terms described above for raster data, except that there is a general assumption that a linear distance of 1 in a 3D mesh coordinate system is equivalent to a linear distance of 1 in some real-world unit (e.g., 1 millimeter, 1 meter, 1 kilometer, etc.). If this is not a valid assumption, a scaling factor would also need to be supplied. In other words if the coordinate system of mesh is such that 4.13 units is equal to 1 millimeter, then a scaling factor term of “1:4.13” and a units term of “mm” would need to be supplied. **We do not recommend a “scaling factor” term at this time and instead strongly recommend that data contributors pre-scale their meshes to accurate, real world units before storing them.**

While Darwin Core has a “units” term (dwc:measurementUnit), this term is meant to be used in combination with dwc:measurementValue. Therefore we recommend the adoption of the more generically applicable Library of Congress bibframe term “Unit” (see below).

As an aside, it would be possible to compute the average “surface resolution” of a mesh as a density (e.g., “average points per mm²”) or scale (e.g., “average point spacing” or “average polygon area”). However, unlike pixel density/scale, these quantities can be derived at any time as long as the point cloud coordinates and coordinate system units have been made available.

URI, Title & Definitions:

bf:unit: “Units in which a value is expressed” (<http://id.loc.gov/ontologies/bibframe/unit>). Additional usage notes: This term shall be used to specify the units of numbers entered into ac:PixelXSpacing, ac:PixelYSpacing, ac:PixelZSpacing for volumetric raster data. When used for mesh and point cloud files, it shall be used to indicate the units of the coordinate system specifying the position of the points and vertices described in the file.

ac:boundingBox: A string of 3 comma separated numbers that give the minimum bounding box of mesh or volume file (e.g., length,width,height). Units for these numbers should be specified with “bf:unit”. This provides a human readable metadata record for the real world size of the scanned object.

III. Terms communicating dimensions or pixel/data counts of image

1. *Raster data*

Dimensions of 3D raster images (which are most commonly represented as a collection of 2D image files) can be described using a series of terms which count pixels along

Audubon Core Enhancement for 3D Imagery

axes (for X, Y, and sometimes Z axes). In some cases (CT and MRI slice series), the number of files in the image collection is equivalent to the dimensions of a “Z axis”. In other cases (CT projection series and photographic collections underlying photogrammetry models), the image collection conveys the third dimension not by its alignment along a Z axis, but by presenting different views of a 3D scene (at different angles and distances from the scene) in 3D space.

Pixel count dimensions along X and Y axes can be expressed using standard 2D image metadata terms for image height and width. Currently, Audubon Core borrows two Exif fields for this purpose: PixelXDimension and PixelYDimension. The Exif schema (see http://cipa.jp/std/documents/download_e.html?DC-008-Translation-2019-E) explicitly defines these fields as only to be used for compressed (e.g., JPEG) image data and that they should not be used for any uncompressed image data. The more general terms recommended in the Exif schema for this purpose are ImageWidth and ImageLength. Therefore (below) we recommend adding these terms to Audubon Core, as they are more generally applicable to uncompressed imagery.

However, a new term is needed for describing the third dimension. We recommend reporting the number of files in image collection and/or the number of Z axis pixels. For reconstructed CT and MRI image slice stack (as in tiff series or dicom series), the number of files should be equivalent to pixel count along the Z axis. If we were just considering image slice stacks here, then by analogy with exif terms like ImageHeight and ImageWidth, one might propose ac:ImageDepth, however such a term would not apply well to CT projection series (whether collected through conventional or helical scanners) or to photograph collections underlying photogrammetry models.

DICOM terms such as ImagesInAcquisition (0020,1003)” or ImagesInSeries (0020,1002) would seem perhaps appropriate, but they are overly specific in referencing the acquisition event itself, when in fact, all that is necessary to know is the number of images (“instances”) comprising the multi-file raw (or first derivative) dataset that has been archived. Sometimes this could be equivalent to the number of images in acquisition, but often it is not. For example in CT, the number of images in acquisition is set by the number of projections and frame averaging, whereas the number of images in a reconstructed CT stack is usually related to the vertical span of scan as well as pixel resolution and density, among other things. Another option could be DICOM’s NumberOfSlices (0054,0081), however this seems like a term that is specific to NM (Nuclear Medicine) imaging that refers to images separated by time increments and again, it would not apply well to x-ray projection series or photograph collections. The DICOM standard term numberOfSeriesRelatedInstances (0020,1209) is a query-level term already used for the purpose of indicating the number of slices for CT modalities and could be appropriate to adopt here since it does not intrinsically assume that the referenced images are in series. It could apply to both raw (projection) and derived (slice) volumetric raster data, as well as photograph collections. However, we think direct

Audubon Core Enhancement for 3D Imagery

adoption of this term would lead to confusion when applied to imaging modalities that are not typically used in medicine (like photogrammetry and laser scans). Therefore we propose a new term `ac:numberOfImages` to use in place of `numberOfSeriesRelatedInstances`.

Title & Definitions:

exif:imageWidth: The number of columns of image data, equal to the number of pixels per row. In JPEG compressed data a JPEG marker is used instead of this tag.

(<http://www.w3.org/2003/12/exif/ns#imageWidth>) *Equivalent to `dicom:columns` (0028, 0011).

exif:imageLength: The number of rows of image data. In JPEG compressed data a JPEG marker is used. (<http://www.w3.org/2003/12/exif/ns#imageLength>) *Equivalent to `dicom:rows` (0028, 0010).

ac:numberOfImages: The number of images or z-axis pixels that provide information on a 3D scene. Inspired by `dicom:numberOfSeriesRelatedInstances` (0020,1209).

Additional usage notes for “`numberOfImages`”: This term shall be used to specify the number of files or slices in a projection series (CT), reconstructed image series (CT, MRI), or that were collected in a photogrammetry session. In cases where volume raster data is represented as a single file (Nifti, NRRD, 3D tiff, dicom Multi-files) it should be used to give the number of Z axis pixels. Other possible uses include any case where each file in a collection represents a part or a view of a single or multiple focal objects that together can be used to understand the 3D nature of the focal object(s). For instance, raw data from laser scans and structured light scans often include a number of partial meshes that represent different parts of the scanned object which need to be processed through alignment and merging before rendering as a full 3D object.

2. *Mesh/point cloud data*

The closest equivalent of raster data’s pixel count dimensions for mesh data would be counts of the discrete data elements comprising the mesh. For polygonal meshes, this would include both a count of the points/vertices positioned in the 3D space coordinate system and a count of the polygonal faces connecting vertices and defining mesh surfaces. For point clouds, this would include only the point/vertex count described previously.

URI, Title & Definition:

ac:pointCount: Number of vertices with X, Y, Z coordinates

ac:faceCount: Number of triangles or polygons with a unique set of three or more points as their vertices.

ac:numberOfImages: The number of images or z-axis pixels that provide information on a 3D scene

IV. Terms communicating additional information about object

1. *Raster data*

Volumetric data from CT scans consist of measurements of X-ray attenuation at each voxel. These values are dependent upon the wavelength and intensity of the X-rays, the characteristics of the object being scanned, the sensitivity of the detector, the exposure time, and any corrections or calibrations applied. While CT reconstruction software may represent these measurements as 32-bit floating point values, CT data are often distributed as 16-bit (unsigned integer) TIFF files, thus a scaling operation is required to map calibrated or uncalibrated floating point values to the integers from 0 to 65535. We propose using three terms from DICOM to capture information about the mapping and allow recalculation of floating point values from the TIFF files. The terms assume a linear transformation of the form:

$$\text{Output units} = m * SV + b$$

where m is the RescaleSlope, SV is the stored value, and b is the RescaleIntercept. The RescaleType term specifies the units of “Output unit” above (e.g., HU for Hounsfield Units, Z_EFF for effective atomic number, US for unspecified, etc.).

DICOM: rescaleIntercept:

<https://dicom.innolitics.com/ciods/ct-image/ct-image/00281052>

DICOM: rescaleSlope: <https://dicom.innolitics.com/ciods/ct-image/ct-image/00281053>

DICOM: rescaleType: <https://dicom.innolitics.com/ciods/ct-image/ct-image/00281054>

To correctly present volumetric data, the location of the xyz origin in the volume must be known. Assumptions regarding origin location can lead to inverted or ‘mirrored’ representation of the data. Typically, the xyz origin will be located at one of the corners of the volume (a rectangular prism). We propose a new term to indicate the location of the volume’s xyz origin in a simple manner when data are distributed as a series of TIFF images. The term consists of three values separated by pipes. The first value is the location of the x origin in the TIFF images (i.e., “left” or “right”), the second value is the location of the y origin in the TIFF images (i.e., “top” or “bottom”), the third value is the image in the series that nearest the z origin (i.e., “first” or “last”). An example of a common case would be “left|top|first”.

ac:volumeOrigin: three ordered values separated by pipes giving the location of the x, y, and z origins in the data. The first value can only be “left” or “right”, the second value can only be “top” or “bottom”, and the third value can only be “first” or “last”.

2. *Mesh data*

Audubon Core Enhancement for 3D Imagery

Our consideration of mesh data here does not comprise an entire scene model, but only encompasses data that are directly applied to the mesh file being described.

Scene-based formats (like .wrl, x3d, and glTF) can get quite intricate, specifying lighting direction, animations, transformations, or relationships in space between multiple objects. The following fields do not account for these, and instead seek to describe properties or information intrinsic to mesh or point geometry. In addition to the 3D coordinate positions of points/vertices and polygonal faces created by the edges connecting them, mesh points and/or faces can have additional types of information encoded alongside these primary properties. These additional attributes can include point or face color, tangential normal vector coordinates, or UV coordinates. UV or texture coordinates comprise an additional 2D coordinate position that can be used to associate meshes with a 2D space, so that visual display of meshes can be enriched with 2D texture maps such as depth maps, bump maps, specularity, and other visual or quantitative properties. For many mesh file formats (e.g., OBJ), these texture maps or other files (such as an MTL materials definition file) are required to associate and achieve an enriched display.

Title & Definitions:

ac:textureMapType: A list (concatenated and separated) of types of texture maps associated with the 3D resource.

Notes: Recommended best practice is to separate the values in a list with space vertical bar space (|).

Example: color-diffuse | specular-gloss

ac:uvCoordinatesBoolean: Indicates whether mesh file has UV coordinates that map it to one or more materials (texture files). Values will be "True" / "False".

ac:vertexColorBoolean: Indicates whether each vertex with a unique x,y,z coordinate also has an associated color specified. Values will be "True" / "False".

ac:normalsBoolean: Indicates whether each vertex or face also has an associated normal vector specified. Values will be "True" / "False".

AC 3D Terms - Implementation Experience Report

The 3D Task Group approached implementation testing by comparing volunteered datasets and data models from MorphoSource (<https://www.morphosource.org>), Smithsonian's Packrat system (<https://smithsonian.github.io/dpo-packrat/>), and University of Michigan Deep Blue Data repository (<https://deepblue.lib.umich.edu/data>). Ease of data entry could not be assessed in all cases, but publicly documenting the purpose of the proposed `ac:Digital3DResource` term in itself facilitates repositories and platforms in need of guidance on "Digital3DResource" usage prior to adding it to their controlled vocabularies for `dcterms:type` or similar terms.

Manual Data Entry

Dedicated data-entry fields for all terms exist in MorphoSource, and many are also present in Packrat. Supplementary sidecar files are necessary in Deep Blue.

Data Findability

Terms were only searchable within MorphoSource, but this highlighted their utility in distinguishing among 3D resources that were generated via different methods (modalities).

- **Digital3DResource:** In repositories encompassing a broader range of 3D as well as non-3D resource types (e.g. DataCite, Deep Blue), testers highlighted the difficulty posed by not having a way to distinguish between 3D- and other resources. The new 'Digital3DResource' term and its defined usage offers a way to address these limitations for data aggregators and data providers whose record schemes already include `dcterms:type` or a term which otherwise maps to it.

Supplementary Examples:

- Metadata Mapping for New terms:
https://docs.google.com/spreadsheets/d/1IPHoN-5RWEQRewUnPT1O_mqJN63WBucOH4Gg4ZemS7Y
- [Link to DataCite discussion/s & proposal/s?]